EXPERIMENT 9 - PYTHON CODE

```
# EXPERIMENT 9 - Forward Chaining
facts = {"fever", "cough", "high_temperature", "age_over_65"}
rules = [
    (["fever", "cough"], "respiratory_symptoms"),
    (["respiratory_symptoms", "high_temperature"], "suspected_infection"),
    (["suspected_infection", "age_over_65"], "priority_consultation")
]

print("FORWARD CHAINING USING FACTS AND RULES")
print("=" * 45)
print("\nStep 1: Initial Facts")
for fact in sorted(facts):
    print("  -", fact)

print("\nStep 2: Apply Rules")
changed = True
round_no = 1
inferred = []
while changed:
    changed = False
    print(f"\nInference Round {round_no}:")
    for conditions, conclusion in rules:
        print(f"\n  Checking: IF {conditions} THEN {conclusion}")
        if all(item in facts for item in conditions):
            if conclusion not in facts:
                facts.add(conclusion)
                inferred.append(conclusion)
                changed = True
                print("  Conditions satisfied.")
                print("  New fact inferred:", conclusion)
            else:
                print("  Conclusion already exists.")
        else:
            missing = [item for item in conditions if item not in facts]
            print("  Rule not applied. Missing:", ", ".join(missing))
    if not changed:
        print("\n  No more new facts can be inferred.")
    round_no += 1

print("\n--- Final Result ---")
print("Newly inferred facts:", inferred)
print("\nFinal set of facts:")
for fact in sorted(facts):
    print("  -", fact)
if "priority_consultation" in facts:
    print("\nConclusion: Patient qualifies for a priority consultation.")
else:
    print("\nConclusion: Priority consultation criteria not established.")
print("Total new facts inferred:", len(inferred))
```